AITU OUTBREAK

Interactive campus rescue with dynamic lighting

Astana IT University
Computer Graphics Fundamentals · Academic year 2026–2027

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Groups supplied: IT-2504 / IT-2503

Project scope and current submission status

The program combines a playable three-block stealth rescue with a perspective inspection view of the same third-floor map. The player finds personal equipment, repairs backup power and distributes a gas neutralizer through campus ventilation. The inspection view exposes per-fragment lighting immediately at startup and supports a visible on/off comparison.

Working topic: AITU OUTBREAK with dynamic lighting. The assigned topic number and exact instructor-approved title were not supplied. The requirements refer to a Section 6 topic checklist, but that section is absent from the supplied document. This report verifies common requirements and does not claim approval of an invented topic.



Figure 1. Perspective campus inspection captured from the submitted executable. The light toggle, team, groups and controls are visible.

Repository: github.com/shaqx77/AITU-OUTBREAK

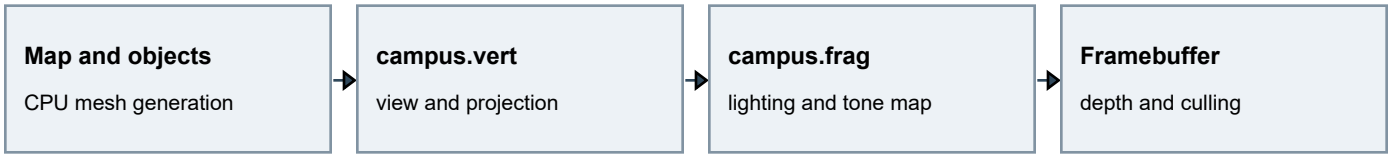
Feature checklist and evidence

Common requirement	Status	Implementation or evidence
OpenGL 3.3+ core without a full engine	Done	GLFW core context; custom C++ renderers.
Perspective camera and resize-safe aspect	Done	CampusView: perspective(48 degrees, width/height); lookAt; orbit and zoom input.
Programmable non-flat shader pair	Done	campus.vert and campus.frag implement diffuse, specular and distance attenuation.
Depth test and consistent face culling	Done	GL_LESS and CCW back-face culling on opaque 3D geometry. UI/2D layers are intentionally exempt.
Movable light and headline toggle	Done	Automatic light orbit; J/L move its angle; B toggles lighting with visible state.
On-screen help and pause	Done	F1/H help; P freezes simulation and animation.
Resources and errors	Done	External shader files, texture checks, FBO completeness, stderr and error.log.
README and reproducible CMake path	Done	Bundled dependencies; Windows x64 build and CTest commands.
Start, goal, failure and sustained play	Done	Three blocks, patrols, objectives, retry checkpoint and rescue ending.
Team and technique visible at startup	Done	Inspection is the startup scene; team/group and lighting state appear on screen.
Diagram, coded math, images and FPS	Done	Pages 3–5; screenshots from this executable and measured on/off table.
Individual contribution ownership	Partial	Proposed module assignments on page 5; actual work and groups require team confirmation.
60–120 second video backup	Done locally	docs/demo.mp4: 60 seconds of actual rendered program frames with a scripted demo path.
YouTube or Google Drive link	Pending	The external upload and access check must be completed before submission.
Assigned topic and Section 6 checklist	Unverified	No topic number, approved title or Section 6 must-implement list available.
Individual shader understanding	Unverified	Each member must perform a live edit and answer questions; tests cannot certify this.

The exact topic-specific checklist cannot be copied or marked complete until the missing topic sheet is provided. No claim is made that bloom, shadow mapping, SSAO, POM, cubemaps or PBR are implemented.

Graphics pipeline and coded mathematics

The inspection view is a forward-rendered 3D view of the tile map. The model is already expressed in world coordinates, so its model matrix is the identity. Positions and per-face normals are batched into a VAO/VBO. The view matrix uses an orbit camera, while projection uses the current framebuffer aspect ratio.



Position / normal / albedo → clip position → per-fragment illumination → UI overlay

2D game: sprite color FBO → lighting and visibility composition → UI → window

Equations implemented in campus shaders

```
clip = projection * view * vec4(worldPosition, 1)
L = normalize(lightPosition - worldPosition)
V = normalize(eye - worldPosition)
H = normalize(L + V)
diffuse = max(dot(N, L), 0)
specular = diffuse > 0 ? max(dot(N, H), 0)^32 : 0
a(d) = 1 / (1 + 0.025*d + 0.002*d*d)
```

```
fill = 0.22 + 0.20 * max(dot(N, normalize(-0.3,1,0.4)), 0)
C = albedo*fill + a(d)*(2.8*albedo*diffuse
+ 0.55*(1.0,0.83,0.58)*specular)
display = (C / (C + 1))^(1 / 2.2)
```

N is the normalized world-space surface normal. The half-vector term creates a Blinn-style highlight. The specular term is disabled on the back-lit side. B selects a fixed albedo baseline instead of the lighting expression. Tone mapping and gamma conversion are applied to both comparison branches.

Lighting in the 2D game

Sprites are drawn to a 640×360 nearest-filtered color texture. A second pass applies lamp attenuation, circular visibility, occupied-room lighting, emergency red pulses, vignette and scanline modulation. The UI is drawn afterward. Alpha sprite/UI layers use painter order and blending, so depth testing and face culling are disabled there. The 3D lighting does not calculate light occlusion or cast shadows from walls.

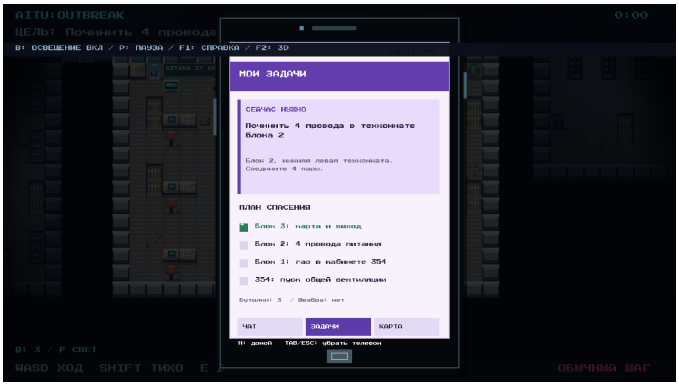
Controls and reproducible demonstration

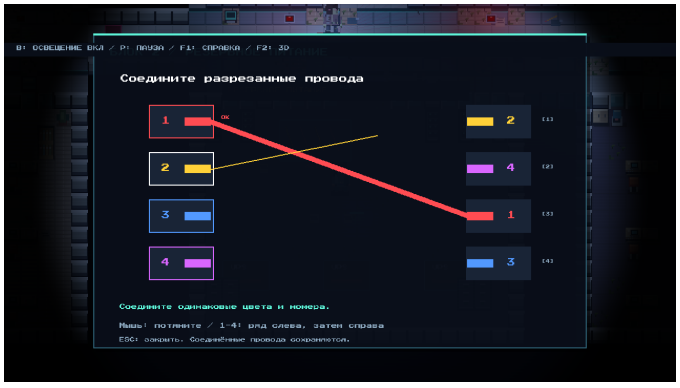
Context	Keys	Purpose
Global	F1/H; F2; B; P; F11	Help; 2D/3D; lighting; animation pause; fullscreen
3D inspection	WASD; Z/X; J/L; F3	Orbit; zoom; move light; whole-floor view
Game	WASD; Shift; E; G/Space	Walk; quiet movement; interaction; temporary stun
Game	Q/right mouse; Tab; Escape/R	Throw; phone; pause/checkpoint retry
Phone	1/2/3; A/D; H; M	Apps; tabs; Home; map overview
Wire panel	Mouse drag or two row keys	Connect a left wire to its matching right socket

Best graphics shot: launch in the default inspection view, leave lighting on, use J/L to move the highlight across the floor, then press B twice. F3 shows the full map. Resize the window to demonstrate perspective aspect handling. The scene appears without first navigating a menu.









Figures 2–5. Lighting enabled and disabled at the same camera position; current phone task; the four-wire backup-power panel. All images are captures from this build.

Game route: 361 computer → locker → service keys → lecture recording and card → block-3 reader → block-2 power repair → room 354 → neutralizer → ventilation. The final camera reveals the whole floor, gas disperses before the cure, then the teacher approaches the player in 354.

Performance and validation

1280×720 inspection	Frames	Total ms	FPS
lighting_off	180	547.676	328.661
lighting_on	180	566.098	317.966

Test device: Intel(R) UHD Graphics. OpenGL version: 3.3.0 - Build 32.0.101.6790. Each mode uses 30 warm-up frames followed by 180 completed frames. glfwSwapInterval(0) removes the normal presentation cap; glFinish includes CPU submission and completed GPU work. This is not a GPU-only timer query. Local results exceed the 30 FPS target; another lab PC must be measured separately.

The automated test checks shader loading, perspective rendering and square-aspect rendering without GL errors, orbit input, lighting toggle, animation pause and help. Gameplay checks include line-of-sight walls, staggered patrols, checkpoints, password/PIN, phone choices, key selection, timed card swipe, wrong wire rejection and partial repair persistence, assisted stun, complete gas coverage, dispersal before cure and teacher arrival.

Proposed module ownership

Team member	Named responsibility
Yerassyl Altynbayev	Rendering and shaders: CampusView.hpp and assets/shaders
Nurali Amangeldi	Gameplay and interaction: main.cpp state machine, AI, puzzles
Daniyar Urazbayev	Map/assets, report, screenshots and backup demonstration

These are proposed review and ownership assignments, not a verified record of individual authorship. The team must confirm actual contributions and which group each member belongs to. Every member must be able to explain and modify the headline shader.

Implementation assistance and personal defence

The implementation received substantial generative assistance, including game logic, shader implementation, campus inspection, asset-generation scripts and documentation. Assisted portions are located in src/main.cpp, src/CampusView.hpp, src/DemoTools.hpp, assets/shaders and tools. Their inclusion is disclosed; this report does not assert independent team authorship of generated portions. Team members must review these files and demonstrate understanding at the defence.

Suggested live modification: change the specular exponent from 32 to 8 and explain the wider highlight; change the distance coefficient and predict the falloff. The included defence guide provides a five-minute presentation outline and further questions.

References and asset provenance

AITU OUTBREAK contributors. (2026). *AITU OUTBREAK* [Computer software]. https://github.com/shaqx77/AITU-OUTBREAK

Astana IT University. (2026). *Computer Graphics Fundamentals final project student requirements 2026–2027* [Unpublished course handout supplied by the team].

de Vries, J. (n.d.). *Basic lighting*. LearnOpenGL. https://learnopengl.com/Lighting/Basic-Lighting

GLFW contributors. (2024). *GLFW 3.4 documentation*. https://www.glfw.org/docs/3.4/

Khronos Group. (2010). *The OpenGL graphics system specification version 3.3 core profile*. https://registry.khronos.org/OpenGL/specs/gl/glspec33.core.pdf

G-Truc Creation. (n.d.). *OpenGL Mathematics* [Software]. https://github.com/g-truc/glm

Barrett, S., et al. (n.d.). *stb* [Software]. https://github.com/nothings/stb

Venn, D. R. (n.d.). *miniaudio* [Software]. https://github.com/mackron/miniaudio

Lohmann, N. (n.d.). *JSON for Modern C++* [Software]. https://github.com/nlohmann/json

CodeMan38. (n.d.). *Press Start 2P* [Font, SIL OFL 1.1]. https://fonts.google.com/specimen/Press+Start+2P

rubberduck. (n.d.). *80 CC0 creature SFX* [Audio pack, CC0]. OpenGameArt. https://opengameart.org/content/80-cc0-creature-sfx

Kenney. (n.d.). *Voiceover pack* [Audio pack, CC0; historical generator sources]. https://kenney.nl/assets/voiceover-pack

NoisyRedFox. (2024, June 20). *Hm Voice* [Audio pack, CC0; historical generator sources]. Freesound. https://freesound.org/people/NoisyRedFox/packs/41303/

GLAD loader provenance and upstream license notices are retained in THIRD_PARTY.md and vendor. Current dialogue blips, music and crowd cheers are locally synthesized. The pixel environment and portraits were prepared for this project with procedural and generative assistance. The 3D geometry is generated from the project map. No paid asset pack is required.

Submission actions still required

Confirm the assigned topic number and exact title; supply the missing topic-specific checklist. Confirm the individual contribution table. Publish docs/demo.mp4 to YouTube or Google Drive, test access and insert that URL here. External video URL: **not yet provided**. The public repository copy is a convenience, not the required platform link.